02. RESEARCH METHODOLOGY

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1. INTRODUCTION TO THE METHODOLOGY

Research methodology constitutes the set of procedures, techniques, and tools that guide the collection, analysis, and interpretation of data to answer the formulated research questions. Within the design science paradigm, methodology acquires a particular dimension: it does not only seek to describe or explain a phenomenon, but to build and evaluate an artifact that solves a practical problem (Hevner et al., 2004).

As specialized literature points out, "research in design sciences is concerned with how things ought to be, that is, with devising technological artifacts to achieve goals" (Simon, 1996). In this project, the central artifact is an intelligent digital twin model with cognitive learning for predicting and optimizing costs, schedules, and risks in construction projects. The methodology must guarantee both the technical quality of the artifact and its practical utility in the Peruvian context.

To guide the selection and justification of methods, the EDFCV framework is applied. This is a tool developed in the course that allows for systematic evaluation of the adequacy of each methodological approach. This framework considers five fundamental dimensions, detailed below, which will be applied to each phase of the project.

2. THE EDFCV FRAMEWORK: FOUNDATIONS AND APPLICATION

The EDFCV framework is a methodological decision-making tool that enables evaluation of a research approach's adequacy based on five criteria:

Criterion	Meaning	Guiding Question
E	Epistemological Fit	Does the chosen method align with my research paradigm and my view of what constitutes valid knowledge?
D	Data Availability	Can I access the data needed to implement this method within the available time?
F	Feasibility	Is it realistic to implement this method considering my resources (time, budget, technical skills, ethical approvals)?
C	Contribution	What type of contribution does this method generate (theoretical knowledge, practical artifact, empirical evidence)?
V	Venue	Are there academic venues (conferences, journals) where this type of research is valued and publishable?

3. APPLICATION OF THE EDFCV FRAMEWORK TO THE PROJECT

3.1. Epistemological Fit

The first criterion, and perhaps the most fundamental, evaluates whether the proposed method aligns with the selected research paradigm. Since this project is framed within design science, the methodology must reflect the principles of this paradigm: artifact construction, utility-based evaluation, and iterative approach.

In this project, the main method is the design, development, and evaluation of a digital twin model with cognitive learning. This approach is epistemologically consistent with design science because:

It builds an artifact: The digital twin model is the tangible outcome of the research, not merely a passive observation of reality. As Simon (1996) points out, design science is concerned with "devising artifacts to achieve goals."

It generates knowledge through construction: As Hevner et al. (2004) state, "knowledge and understanding of a problem domain and its solution are achieved in the construction and application of the designed artifact." It is not just about observing the problem, but about building it, testing it, and learning in the process.

It evaluates by utility: The success of the method is measured by the model's predictive accuracy (MAPE < 12%), its ability to reduce deviations (>30%), and its accessibility for Peruvian medium-sized companies (cost < $15,000).

Justification of alignment: Predicting and optimizing costs, schedules, and risks in construction is not a problem that can be solved solely through observation or theory. It requires building a system that learns patterns from historical data and applies them in a practical context. Design science provides the appropriate framework to address this challenge.

3.2. Data Availability

The second criterion evaluates whether the data needed to implement the method is available. For predicting costs, schedules, and risks in construction, access to historical project datasets is required, including actual costs, actual durations, change orders, and contextual variables.

Identified data sources:

Data from Peruvian construction companies: Contact has been established with 3-5 medium-sized construction companies in Metropolitan Lima that have expressed willingness to share anonymized historical data from their projects (estimated: 40-100 projects).

Public data: There are international public datasets (e.g., data.gov, World Bank Infrastructure Projects) that can complement the analysis.

Government data: The Ministry of Transport and Communications and the Ministry of Housing publish data on public projects that can be utilized.

Synthetic data generation: To complement real data, synthetic data generation techniques that preserve the statistical properties of the original data will be used.

Data availability assessment: Data availability is high for the training and initial validation phases (data from participating companies), but medium for validation in the broader Peruvian context, which will require additional institutional agreements. This aspect will be considered in the timeline planning.

3.3. Feasibility

The third criterion evaluates the practical viability of the method considering resources, time, skills, and ethical approvals.

Time: The system development is estimated at three years, distributed as follows:

Year 1: Literature review, establishment of agreements with companies, collection and structuring of historical data.

Year 2: Design of the digital twin architecture, training and comparison of cognitive learning algorithms.

Year 3: Validation in real projects, results analysis, and thesis writing.

Computational resources: Accessible platforms such as Google Colab (free or paid plan) or institutional resources will be used, which are adequate for training moderate-sized machine learning models.

Technical skills: The researcher has training in machine learning, data processing, and experience in the construction sector, enabling implementation tasks with domain knowledge.

Ethical approvals: Approval from the UNMSM Research Ethics Committee will be required for handling company data. For public data, the corresponding approvals are already in place.

Feasibility assessment: The project is feasible with available resources, although careful planning is required for the validation phase in real projects.

3.4. Contribution

The fourth criterion evaluates the type and scope of the contribution generated by the method.

Expected types of contribution:

Functional artifact: A predictive digital twin model that construction companies can use as a decision-support tool.

Methodological knowledge: A low-cost digital twin architecture adapted to the Peruvian context, including training strategies and explainability mechanisms.

Empirical evidence: Quantitative results on the system's predictive accuracy (MAPE, R^2) and its impact on deviation reduction (>30%).

Implementation guide: Documentation enabling other researchers and professionals to replicate and adapt the system.

Contribution assessment: The contribution is high, as it combines a practical artifact, methodological knowledge, and empirical evidence, all aligned with the design science paradigm.

3.5. Venue (Publication Venues)

The fifth criterion evaluates whether there are academic venues where the research can be published and valued.

Identified venues:

International conferences: NeurIPS, ICML, KDD, IEEE Big Data.

Specialized journals: Automation in Construction, Journal of Construction Engineering and Management, Advanced Engineering Informatics.

Open access journals: Scientific Reports, PLOS ONE, where recent research on machine learning applied to construction has been published.

Venue assessment: The research has a high potential for publication due to the practical relevance of the topic, the currency of cognitive learning techniques, and the focus on a context (Peru/Latin America) with limited prior research.

4. COMPARISON OF CANDIDATE METHODS

Below is a summary table of the evaluation of different methods considered for the project, using the EDFCV framework. Each method is rated on a scale of 1 to 5 (1 = low adequacy, 5 = high adequacy) for each criterion.

Method	E (Epistemological)	D (Data)	F (Feasibility)	C (Contribution)	V (Venue)	Total Score
Digital Twin + ML (selected)	5	4	4	5	5	23
Classical Machine Learning (prediction only)	3	4	5	3	3	18
Qualitative Case Study	2	5	4	2	2	15
Systematic Literature Review	3	5	5	3	4	20
Monte Carlo Simulation	3	3	4	3	3	16

.1. Justification for Selecting the Main Method

The Digital Twin integrated with Machine Learning (cognitive learning) is the method that obtains the highest score in the EDFCV framework (23 points). Below is the detailed justification:

Epistemological alignment (5/5): Developing a digital twin with cognitive learning is a paradigmatic example of design science applied to construction. Building a system that learns from historical data and predicts future behaviors is an artifact creation that solves a practical problem (Hevner et al., 2004).

Data availability (4/5): Multiple data sources exist: Peruvian construction companies have expressed willingness to share historical data; there are international public datasets; and synthetic data can be generated to complement.

Feasibility (4/5): The project is technically feasible with available resources (Google Colab, Python libraries). The main challenge is validation in the Peruvian context, which requires additional arrangements.

Contribution (5/5): The contribution is high across all dimensions: functional artifact, methodological knowledge (low-cost architecture), and empirical evidence.

Publication venues (5/5): The topic is of great interest to conferences and journals specialized in construction and artificial intelligence.

4.2. Comparison with Alternative Methods

Classical Machine Learning (18 points): While classical methods (Random Forest, XGBoost) can achieve acceptable results, they do not capture the complexity of interaction between contextual variables as effectively as an integrated digital twin. Their lesser capacity to simulate "what-if" scenarios limits their contribution.

Qualitative Case Study (15 points): A study based on interviews with project managers could provide valuable information about practical needs, but it does not directly address the technical problem of predicting costs and schedules. Its contribution would be complementary, not central.

Systematic Literature Review (20 points): A systematic review of the literature on digital twins in construction would contribute to theoretical knowledge, but it would not generate a practical artifact that solves the problem. It is a valuable activity, but insufficient as the main method in the design science paradigm.

Monte Carlo Simulation (16 points): Monte Carlo simulation would provide probabilistic estimates, but it would not incorporate learning from historical data or optimization capabilities. Its contribution would be limited compared to the digital twin with ML approach.

5. DETAILED PROJECT METHODOLOGY

Based on the EDFCV framework evaluation, the project methodology is structured in the following phases:

5.1. Phase 1: Literature Review and Problem Definition

Objective: Establish the state of the art in digital twins and machine learning applied to construction project management, and identify research gaps and opportunities.

Activities:

- Systematic search in databases (Scopus, Web of Science, PubMed) for articles published between 2018 and 2026.
- Analysis of digital twin architectures used in construction.
- Identification of available datasets and their characteristics.
- Definition of system requirements (accuracy, interpretability, accessibility).

5.2. Phase 2: Data Collection and Preparation

Objective: Collect, structure, and preprocess historical construction project data for model training and validation.

Activities:

- Establishment of confidentiality agreements with participating construction companies.
- Collection of historical data: actual costs, actual durations, change orders, contextual variables (project type, location, size, etc.).
- Data cleaning and structuring: format standardization, handling of missing values, outlier detection.
- Creation of a structured data repository (unified dataset).
- Division of the dataset into training (70%), validation (15%), and testing (15%).

5.3. Phase 3: Digital Twin Design and Development

Objective: Design, implement, and optimize a digital twin model with cognitive learning for predicting costs, schedules, and risks.

Activities:

- Digital twin architecture design: Definition of the 4 layers (data capture, virtual model, ML engine, visualization).
- ML algorithm selection and training: Comparison of algorithms (XGBoost, CatBoost, Random Forest, LSTM Networks) for cost overrun and delay prediction.
- Cognitive learning incorporation: Implementation of case-based reasoning (CBR) and explainability techniques (SHAP, LIME).
- Prototype development: Implementation of the system in Python using open-source tools.

5.4. Phase 4: Evaluation and Validation

Objective: Evaluate the system's performance on test datasets and in real projects.

Activities:

- Model evaluation on the test set using standard metrics: MAPE, R^2, MAE.
- Comparison of model performance with traditional methods (CPM, EVM).
- Validation in 1-2 ongoing projects (monitoring for 6-8 months).
- Evaluation of model interpretability using SHAP and visualizations.
- Analysis of deviation reduction (>30%) and improvement in decision-making.

5.5. Phase 5: Documentation and Communication

Objective: Document the system and communicate results to the academic and professional community.

Activities:

- Preparation of Model Card and Data Sheet.
- Thesis writing.
- Preparation of articles for publication in indexed journals.
- Development of a practical guide for construction companies.

6. VALIDATION AND METHODOLOGICAL RIGOR

In the design science paradigm, artifact validation is an essential component of methodological rigor. The digital twin model evaluation will be conducted across multiple dimensions:

6.1. Technical Validation

Predictive accuracy: The model's ability to predict cost overruns and delays will be measured using standard metrics (MAPE, R^2, MAE).

Robustness: System performance will be evaluated on different datasets (cross-validation) and under different conditions.

Reproducibility: Code, data, and model parameters will be documented to enable other researchers to replicate the results.

6.2. Practical Validation

Deviation reduction: The model's impact on reducing budgetary and schedule deviations (>30%) will be analyzed.

Usability: Ease of use by construction professionals will be evaluated.

Accessibility: Implementation cost will be verified to be less than $15,000.

6.3. Ethical Validation

Transparency: System limitations and situations where it should not be used will be documented.

Fairness: The system will be verified to have no significant biases across different project types or regions.

7. REFERENCES

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